

**In-Course Assessment Examination - I**  
**Department of Physical Science**  
**IT2143 – Visual Programming (Practical)**  
**First Semester**

**Time: 3 hours**

**April 2022**

---

**Note:** All your implementations will be subject to plagiarism checking. If two implementations have the same copied code completely or partially, then both will have 0 or reduced marks respectively. Using online resources is allowed, except directly copying from websites.

Create a folder mentioning your registration number (e.g: 2018ICTXX) and save all your files inside that folder. Compress the folder and submit to Google Classroom assessment submission link before the deadline.

*Create an application to retrieve personal details of a Sri Lankan citizen from a given NIC number.*

*Old NIC numbers contain 10 characters. The first two digits represent the birth year (For 962311684V year is 1996). New NIC numbers contain 12 characters. The first four digits represent the birth year (For 19962311684V year is 1996). The next three digits (231) represent birth month and day. It's the number of total days from January 1<sup>st</sup> to the birthday. Also, these three digits represent the gender. If it is less than 500, the person is Male, otherwise the person is Female. For females, an additional 500 has been added to the number of days from January 1<sup>st</sup> to the birthday. The last character is the letter 'V' or 'X'. 'V' is for persons who are eligible to vote. 'X' is for persons who are not eligible to vote.*

*Use Visual Studio with C# and follow the given instructions below.*

**(Q1) NIC Text Box**

Create a 'Windows Forms Control Library' and Implement a Custom Control Textbox which allows user to enter numbers and uppercase letters 'V' or 'X' only. It should further allow the user to use the 'Backspace' key. It should not allow any other characters to enter. It should have any difference in appearance to be identified separately.

[20%]

**(Q2) NIC Operations Class Library**

Create a 'Class Library' which has all the functions necessary to retrieve a person's,

- Gender
- Date of Birth in format of 1987.07.25
- Vote eligibility

It should be implemented to support old (962311684V) and new (19962311684V) NIC number types. Consider the number of days in February as 29. It should be built as a Dynamic Link Library (DLL) file.

[50%]

### (Q3) NIC Personal Details Application

Create a 'Windows Forms Application' making use of the custom control textbox created in Q1 and the DLL file in Q2, applying OOP concepts. This application is to display personal information as mentioned in Q2. Consider the following functionalities.

- Initially, 'Next Person' button should be disabled and cursor should be focused in input text box.
- For the input textbox, created custom control textbox should be used with relevant restrictions.
- User should enter the NIC number and then pressing 'Enter' key should display the details in read-only text boxes.
- Then 'Next Person' button should be enabled and focused.
- When clicking on 'Next Person' button or pressing 'Enter' key should make all text boxes empty and switch to next person and allow user to enter the next person's NIC number.
- In such a way that the user should be able to retrieve personal information of any number of persons, entering NIC numbers one by one and pressing 'Enter' key.
- If user has entered an NIC number which does not have number of characters according to new or old NIC number format, then system should inform the user as 'Invalid NIC!' via a MessageBox.

[30%]

Form will be as follows:

The application window is titled "NIC - Personal Details". It contains the following elements:

- NIC Number:** A custom control textbox with a yellow background and a blue border.
- Next Person:** A button that is disabled (gray) initially and becomes enabled (blue) after a valid NIC number is entered.
- Gender:** A read-only text box.
- Date of Birth:** A read-only text box.
- Vote Eligibility:** A read-only text box.

The four screenshots illustrate the application's behavior:

- Initial State:** The "NIC Number" field is empty, and the "Next Person" button is disabled.
- Valid NIC Entry:** The user enters "962311684V". The "Next Person" button becomes enabled, and the details (Gender: Male, Date of Birth: 1996.08.18, Vote Eligibility: Eligible) are displayed.
- Next Person Click:** The user clicks the "Next Person" button. The "NIC Number" field is cleared, and the "Next Person" button becomes disabled again.
- Invalid NIC Error:** The user enters "964325V". A message box titled "Invalid NIC!" appears with an "OK" button.